Department of Biomedical Engineering
Riphah College of Science and Technology
Faculty of Engineering & Applied Sciences
Riphah International University, Lahore

Course Title : Biomedical Signal Processing
Faculty : Engr. Rana Talha Khalid
Designation : Lab Engineer
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Course Learning Outcomes (CLOs)

CLO No.	Course Learning Outcome (CLO)	Domain & Level	Program Learning Outcomes PLOs
CLO1	Adapt and implement algorithms for processing biomedical signals using MATLAB/Python and evaluate their performance.	Cognitive C3	PLO 5 Tool Usage
CLO2	Analyze and interpret biomedical signals to extract meaningful features that are helpful in diagnosis of different abnormalities using appropriate signal processing techniques.	Cognitive C4	PLO 4 Investigation
CLO3	Prepare a comprehensive laboratory report and effectively communicate the experimental findings, methodology, and conclusions through oral presentation.	Affective– A2	PLO 9 Communication

Domain & Level	Behavior Description
Cognitive C3	The student selects, transfers, and uses data and principles to complete a problem or task with a minimum of direction.
Cognitive C4	The student distinguishes, classifies, and relates the assumptions, hypotheses, evidence, or structure of a statement or question.
Affective – A2	The student actively participates, responds willingly, and shows a positive attitude toward learning activities, particularly in the preparation and completion of laboratory reports.

Mapping of Course Learning Outcomes (CLO) to Program Learning Outcomes (PLO)

Course Code	CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO 10	PLO 11
CSML – 101	CLO 1					X						
	CLO 2				X							
	CLO 3									X		

PLO1	Engineering Knowledge	PLO7	Ethics
PLO2	Problem Analysis	PLO8	Individual and Teamwork
PLO3	Design/Development of Solutions	**PLO9**	**Communication**
PLO4	**Investigation**	PLO10	Project Management
PLO5	**Tool Usage**	PLO11	Lifelong Learning
PLO6	The Engineer and the world		

PLO 4	**Investigation**	An ability to investigate complex engineering problems in a methodical way including literature survey, design and conduct of experiments, analysis and interpretation of experimental data, and synthesis of information to derive valid conclusions.
PLO 5	**Tool Usage**	An ability to create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering activities, with an understanding of the limitations.
PLO 9	**Communication**	An ability to communicate effectively (and inclusively, in some versions) on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

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Program: B.Sc. Biomedical Engineering
Semester: VI

Subject: Biomedical Signal Processing

LIST OF EXPERIMENTS

Experiment No	Experiment Title
1.	R peak detection using Pan-Tompkins Algorithm
2.	Finding Heart Rate Variability (HRV) using Pan-Tompkins Algorithm
3.	Time domain and frequency domain analysis of HRV
4.	Non-linear and statistical analysis of HRV
5.	Approximate Entropy and Sample Entropy computation on HRV signals
6.	**Open-ended Lab:** Develop a full ECG and HRV analysis dashboard
7.	Motion artifact removal from ECG using LMS and comparison with static filters
8.	ECG Feature Extraction for Arrhythmia Analysis (Detect R-peaks, calculate RR intervals, QRS duration, QRS amplitude, and ST-segment level from normal and arrhythmic ECG records; tabulate features for multiple beats)
9.	ECG segmentation and beat template extraction using ensemble averaging
10.	EEG band power extraction, i.e.: Delta, Theta, Alpha, Beta using Welch PSD
11.	EEG spectrogram analysis and sleep stage identification using band power ratios
12.	**Open Ended Lab:** Design a real-time EEG-based attention or drowsiness detection system using band power features
13.	EMG time-domain feature computation, i.e.: RMS, MAV, Zero Crossing Rate, Willison Amplitude
14.	Muscle fatigue detection using median frequency shift and RMS trend analysis of EMG
15.	Wavelet-based denoising of ECG and EMG using DWT soft and hard thresholding
16.	Coherence and cross-correlation analysis between two biosignal channels
17.	**Open ended Lab:** Develop a gesture recognition system using EMG time-domain and frequency-domain features

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Program: B.Sc. Biomedical Engineering **Semester: VI**

Subject: Biomedical Signal Processing **Date**:

Experiment # 1:

R peak detection using Pan-Tompkins Algorithm

Objective:

- To implement the Pan-Tompkins algorithm for accurate detection of R-peaks in ECG signals.
- To evaluate the performance of the Pan-Tompkins algorithm by detecting R-peaks and calculating heart rate from raw ECG recordings.

Name: **SAP ID:**

Targeted CLO(s):

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CLO3	Prepare a comprehensive laboratory report and effectively communicate the experimental findings, methodology, and conclusions through oral presentation.	Affective– A2	PLO 9 Communication

Rubrics:

Need Improvement (Marks 1)	Weak (Marks 2)	Satisfactory (Marks 3)	Good (Marks 4)	Excellent (Marks 5)
Lab Performance				
Attribute: *Algorithm Implementation & Coding Skill*		**CLO 1**	**Marks:**	
Major coding errors, incomplete algorithm	Many mistakes, frequent help needed	Basic implementation with some errors.	Correct implementation with minor issues	Accurate, efficient and well-structured code.
Attribute: *R-peak Detection Performance*		**CLO 2**	**Marks:**	
Very low detection accuracy, many false peaks.	Low sensitivity or high false detections.	Moderate accuracy with noticeable errors	Good detection rate on test signals.	High accuracy, excellent sensitivity and precision.
Lab Reports				
Attribute: *Results, Plots & Analysis*		**CLO 3**	**Marks:**	
Missing or incorrect plots and results.	Incomplete results with poor analysis	Basic plots and simple analysis.	Clear plots with good result analysis	Excellent plots, accurate analysis and comparisons.
Attribute: *Discussion, Conclusion & Presentation*		**CLO 3**	**Marks:**	
No discussion or conclusion, poor format. |	Incomplete discussion, weak conclusion, report disorganized.	Basic discussion and conclusion present.	Clear discussion with reasonable comparison of results, well-organized report.	Outstanding discussion, critical comparison between experimental, simulation & theoretical results, excellent presentation and conclusions.
Remarks:				

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Experiment # 2:

Finding Heart Rate Variability (HRV) using Pan-Tompkins Algorithm

Objective:

- To implement Pan-Tompkins algorithm for R-peak detection and compute Heart Rate Variability (HRV) from ECG signals.

Name: **SAP ID:**

Targeted CLO(s):

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CLO2	Analyze and interpret biomedical signals to extract meaningful features that are helpful in diagnosis of different abnormalities using appropriate signal processing techniques.	Cognitive C4	PLO 4 Investigation
CLO3	Prepare a comprehensive laboratory report and effectively communicate the experimental findings, methodology, and conclusions through oral presentation.	Affective– A2	PLO 9 Communication

Rubrics:

Need Improvement (Marks 1)	Weak (Marks 2)	Satisfactory (Marks 3)	Good (Marks 4)	Excellent (Marks 5)
Lab Performance				
Attribute: *Algorithm Implementation & Coding Skill*		**CLO 1**	**Marks:**	
Major coding errors, incomplete algorithm	Many mistakes, frequent help needed	Basic implementation with some errors.	Correct implementation with minor issues	Accurate, efficient and well-structured code.
Attribute: *HRV Calculation Performance*		**CLO 2**	**Marks:**	
Very low detection accuracy, many false peaks.	Low sensitivity or high false detections.	Moderate accuracy with noticeable errors	Good detection rate on test signals.	High accuracy, excellent sensitivity and precision.
Lab Reports				
Attribute: *Results, Plots & Analysis*		**CLO 3**	**Marks:**	
Missing or incorrect plots and results.	Incomplete results with poor analysis	Basic plots and simple analysis.	Clear plots with good result analysis	Excellent plots, accurate analysis and comparisons.
Attribute: *Discussion, Conclusion & Presentation*		**CLO 3**	**Marks:**	
No discussion or conclusion, poor format. |	Incomplete discussion, weak conclusion, report disorganized.	Basic discussion and conclusion present.	Clear discussion with reasonable comparison of results, well-organized report.	Outstanding discussion, critical comparison between experimental, simulation & theoretical results, excellent presentation and conclusions.
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Experiment # 3:

Time domain and frequency domain analysis of HRV

Objective:

- To perform time-domain analysis of HRV parameters (SDNN, RMSSD, pNN50) from ECG-derived RR intervals.
- To perform frequency-domain analysis of HRV and calculate power in LF, HF, and LF/HF ratio using PSD estimation.

Name: **SAP ID:**

Targeted CLO(s):

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Rubrics:

Need Improvement (Marks 1)	Weak (Marks 2)	Satisfactory (Marks 3)	Good (Marks 4)	Excellent (Marks 5)
Lab Performance				
Attribute: *Time-Domain HRV Analysis*		**CLO 1**		**Marks:**
Incorrect or missing time-domain parameters.	Many errors in SDNN, RMSSD calculation.	Basic parameters calculated with some errors.	Correct time-domain parameters with minor issues	Accurate SDNN, RMSSD, pNN50 with proper method.
Attribute: *Frequency-Domain HRV Analysis*		**CLO 2**		**Marks:**
No frequency analysis or wrong PSD.	Incorrect LF, HF bands and ratio.	Basic PSD done with some errors.	Proper LF, HF power and LF/HF calculated.	Accurate PSD, correct frequency parameters with clear interpretation.
Lab Reports				
Attribute: *Results, Plots & Analysis*		**CLO 3**		**Marks:**
Missing or incorrect plots and results.	Incomplete results with poor analysis	Basic plots and simple analysis.	Clear plots with good analysis of HRV.	Excellent plots, accurate analysis and comparisons.
Attribute: *Discussion, Conclusion & Presentation*		**CLO 3**		**Marks:**
No discussion or conclusion, poor format. |	Incomplete discussion, weak conclusion, report disorganized.	Basic discussion and conclusion present.	Clear discussion with reasonable comparison of results, well-organized report.	Outstanding discussion, critical comparison between experimental, simulation & theoretical results, excellent presentation and conclusions.
Remarks:				

Signature: _____